Retail Sales and Customer Behavior Analysis

An exploratory analysis of retail transaction data to identify customer trends and product performance.

Summary:

This project analyzes retail transaction data to uncover insights into customer purchasing behavior, product category performance, and sales trends. The analysis highlights key revenue drivers, identifies high-value customer segments, and reveals seasonal sales patterns. These insights can help businesses optimize marketing strategies and inventory planning.

PHASE 1: ASK

Business Problem / Objective:

The objective of this project is to analyze retail transaction data to understand customer purchasing behavior, product category performance, and sales trends over time. These insights can help businesses optimize inventory, target customers effectively, and improve sales strategies.

Key Questions

1. Which product categories generate the most revenue?
2. Which customer segments spend the most?
3. How do sales change over time?
4. Which categories contribute most to overall sales?

PHASE 2: PREPARE

Dataset Description:

Dataset source: Kaggle
Dataset name: Retail Sales Dataset

No. of rows: 9
No. of columns: 1000

Data dictionary:

Attribute	Description
Transaction ID	Unique ID for each purchase
Customer ID	Unique identifier for customers
Date	Transaction date
Gender	Customer gender
Age	Customer age
Product Category	Category of product purchased
Quantity	Number of units purchased
Price per Unit	Price of a single unit
Total Amount	Total purchase value

Let us now categorize the attributes:

1. Customer attributes:
    a. Gender

    b. age
2. Product attributes:
    a. Product categories
3. Time:
    a. date
4. Sales metrics
    a. Quantity
    b. Total amount

Metrics Vs Dimensions:

Metrics = numbers we measure
Dimensions = categories we compare

1. Metrics:
    a. Total amount
    b. quantity
2. Dimensions
    a. Product category
    b. Gender
    c. Age
    d. Date

The given dataset is capable of answering the following questions:

Dimension	Metric	Question
Product Category	Total Amount	Which category generates the most revenue?
Gender	Total Amount	Which gender spends more?
Age	Total Amount	Which age group spends the most?
Date	Total Amount	How do sales change over time?
Product Category	Quantity	Which category sells the most units?

PHASE 3: PREPARE

Data Cleaning Steps

1. Checked dataset structure and column names.
2. No duplicate transactions spotted.
3. No missing values as well.
    a. Two empty rows and two empty columns removed.
4. Standardized text formatting for categorical columns
5. Verified revenue calculation (Quantity × Price per Unit)
6. Created additional columns for analysis (Month, Age Group):
    a. Month extracted from date using =TEXT(B2, “MMM”)
    b. Age groups created using:

    =if(F2>=50, "50+", if(F2>=40, "40-50", if(F2>=30, "30-40", if(F2>=18, "18-30", "null"))))

    Hence, the age groups created are 18-30, 30-40, 40-50 and 50+.

7. Prepared dataset for exploratory analysis

PHASE 4: ANALYSE

Here is a summary of what we shall analyse in this project:

Sales Overview

- Total revenue
- Total transactions
- Total quantity sold

Customer Analysis

- Revenue by gender
- Revenue by age group

Product Category Analysis

- Revenue by product category
- Quantity sold by category

Sales overview:

A few KPIs got from the data in hand are as follows:

1. Total Revenue: 456,000
2. Total Transactions: 1,000
3. Total Quantity Sold: 2,514 units
4. Average Order Value: 456
5. Average Items per Order: 2.5

Let us first understand how much in revenue each category contributes:

Product Category	SUM of Total Amount
Beauty	143515
Clothing	155580
Electronics	156905
Grand Total	**456000**

Since revenue needn't necessarily mean quantity sold is the same, let us now see quantity sold under each product category.

Product Category	SUM of Quantity
Beauty	771
Clothing	894
Electronics	849
Grand Total	**2514**

Next, let us see the monthly performance in terms of quantity and revenue:

Month	SUM of Quantity	SUM of Total Amount
May	259	53150
Oct	252	46580
Dec	207	44690
Feb	214	44060
Jan	199	36980
Aug	227	36960
Jun	197	36715
Jul	176	35465
Nov	205	34920
Apr	214	33870
Mar	194	28990
Sep	170	23620
Grand Total	**2514**	**456000**

Finally, it is time to view revenue by month per product category:

	1	2	3	4	5	6	7	8	9	10	11	12

Product Category	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Beauty	13930	14035	10545	11905	12450	10995	16090	9790	6320	15355	9700	12400
Clothing	13125	14560	15065	13940	17455	10170	8250	12455	9975	13315	15200	12070
Electronics	9925	15465	3380	8025	23245	15550	11125	14715	7325	17910	10020	20220
Grand Total	**36980**	**44060**	**28990**	**33870**	**53150**	**36715**	**35465**	**36960**	**23620**	**46580**	**34920**	**44690**

The insights received from these pivot tables can be consolidated as follows:

1. Electronics generated the highest revenue among all product categories, contributing approximately 34.4% of total sales. Clothing followed closely (34.1%), while Beauty products contributed slightly lower (31.5%) but still significant revenue.

2. Clothing products recorded the highest quantity sold, indicating strong customer demand. However, Electronics generated the highest revenue despite lower sales volume, suggesting a higher average price per unit.

3. Sales volume peaked in May and October, indicating periods of high customer demand, while September recorded the lowest sales volume. This suggests seasonal variation in purchasing patterns.

4. Sales performance varies significantly across months. May recorded the highest sales in both revenue and quantity, indicating strong demand. However, months like February and December generated high revenue despite moderate sales volume, suggesting purchases of higher-value products. September showed the lowest performance in both metrics, indicating a dip in customer demand.

These insights have been consolidated into a single dashboard in Phase 5: Share.

Customer analysis:

Under customer analysis, we shall see how each gender and age group adds to the revenue and quantity. After that, we shall also see which product categories are preferred by which gender and age group.

First, let us understand the number of people we are looking at:

Gender	COUNTUNIQUE of Customer ID
Female	510
Male	490
Grand Total	**1000**

Age group	COUNTUNIQUE of Customer ID
18-30	251
30-40	191
40-50	222
50+	336
Grand Total	**1000**

Next, let us see the transactional value:

Gender	SUM of Total Amount
Female	232840
Male	223160
Grand Total	**456000**

Age group	SUM of Total Amount
18-30	123155

30-40	96325
40-50	93365
50+	143155
Grand Total	**456000**

Let us next see the quantity of items purchased:

Gender	SUM of Quantity
Female	1298
Male	1216
Grand Total	**2514**

Age group	SUM of Quantity
18-30	625
30-40	509
40-50	553
50+	827
Grand Total	**2514**

Let us now see the average amount spent by each customer base:

Gender	AVERAGE of Total Amount
Female	456.5
Male	455.4
Grand Total	**456**

Age group	AVERAGE of Total Amount
18-30	490.7
30-40	504.3
40-50	420.6
50+	426.1
Grand Total	**456**

Let us now understand the product preference of each customer segment:

SUM of Total Amount	*Product Category*			
Age group	Beauty	Clothing	Electronics	Grand Total
18-30	41400	46150	35605	123155
30-40	33385	30340	32600	96325
40-50	34645	32020	26700	93365
50+	34085	47070	62000	143155
Grand Total	**143515**	**155580**	**156905**	**456000**

SUM of Total Amount	*Product Category*			
Gender	Beauty	Clothing	Electronics	Grand Total
Female	74830	81275	76735	232840
Male	68685	74305	80170	223160
Grand Total	**143515**	**155580**	**156905**	**456000**

Product Category Analysis

Questions to be answered:

1. Which category generates the most revenue?
2. Is high revenue coming from high price or high volume?
3. Which category has low performance? Why?
4. Are there category preferences by age group?
5. Where is the business underperforming?

Let us start off with the quantity of revenue generated per product category.

Product Category	SUM of Total Amount
Beauty	143515
Clothing	155580
Electronics	156905
Grand Total	**456000**

Next, let us see quantity of products sold per category:

Product Category	SUM of Quantity
Beauty	771
Clothing	894
Electronics	849
Grand Total	**2514**

Next, average amount spent per product category:

Product Category	AVERAGE of Total Amount
Beauty	467.48
Clothing	443.25
Electronics	458.79
Grand Total	**456**

Product category preferences by different age groups:

SUM of Quantity	*Product Category*			
Age group	Beauty	Clothing	Electronics	Grand Total
18-30	215	214	196	625
30-40	170	163	176	509
40-50	170	219	164	553
50+	216	298	313	827
Grand Total	**771**	**894**	**849**	**2514**

A summary of our findings:

Category	**Revenue**	**Volume**	**Type**
Electronics	High	Medium	Premium
Clothing	Medium	High	Volume Driver
Beauty	Medium	Medium	Balanced

PHASE 5: SHARE

Sales overview:

The insights received includes:

1. Electronics generated the highest revenue among all product categories, contributing approximately 34.4% of total sales. Clothing followed closely (34.1%), while Beauty products contributed slightly lower (31.5%) but still significant revenue.
2. Clothing products recorded the highest quantity sold, indicating strong customer demand. However, Electronics generated the highest revenue despite lower sales volume, suggesting a higher average price per unit.
3. Sales volume peaked in May and October, indicating periods of high customer demand, while September recorded the lowest sales volume. This suggests seasonal variation in purchasing patterns.
4. Sales performance varies significantly across months. May recorded the highest sales in both revenue and quantity, indicating strong demand. However, months like February and December generated high revenue despite moderate sales volume, suggesting purchases of higher-value products. September showed the lowest performance in both metrics, indicating a dip in customer demand.

Let us consolidate the above insights into a single dashboard:

Dashboard 1: Sales overview

Customer Insights

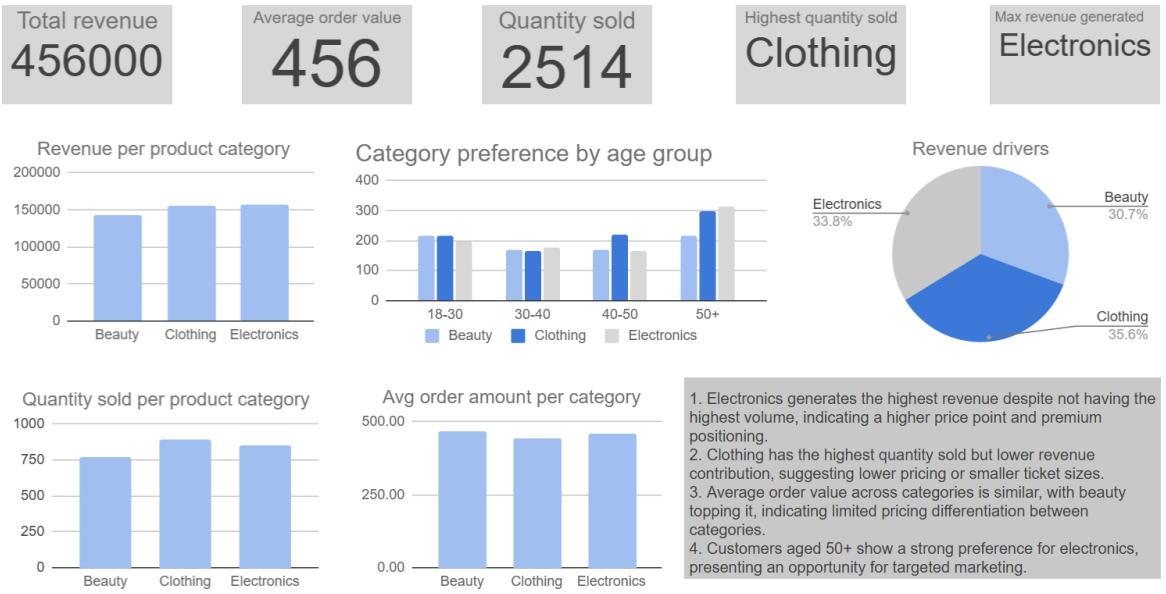
1. The customer base is evenly distributed by gender, with a slight female majority.
2. The 50+ age group represents the largest and most valuable segment, contributing the highest revenue and purchase volume, while also showing a strong preference for Electronics.
3. Customers aged 30–40 have the highest average spend per transaction, indicating higher-value purchases despite lower overall volume.
4. Female customers demonstrate marginally higher engagement, generating more revenue and purchasing more items than male customers. Gender-based preferences show that females lean toward clothing, while males contribute more to electronics sales.
5. Younger customers (18–30) contribute significantly but show lower overall impact compared to older segments.They prefer clothing and beauty products, indicating a lifestyle-driven purchasing pattern.

Let us see how these insights translates into a dashboard:

Dashboard 2: Customer overview

Product category analysis:

1. Electronics generates the highest revenue despite not having the highest volume, indicating a higher price point and premium positioning.
2. Clothing has the highest quantity sold but lower revenue contribution, suggesting lower pricing or smaller ticket sizes.
3. Average order value across categories is similar, with beauty topping it, indicating limited pricing differentiation between categories.
4. Customers aged 50+ show a strong preference for electronics, presenting an opportunity for targeted marketing.



Dashboard 3: Product category analysis

PHASE 6: ACT

Recommendations based on sales overview:

1. The business can focus on promoting Electronics further to maximize revenue, while also maintaining strong inventory levels across all categories since demand is evenly distributed.

2. The business can use Clothing products to drive customer traffic due to high demand, while focusing on Electronics to maximize revenue through higher-value sales.

3. The business should investigate high-revenue months like February and December to understand which products drive higher-value sales. Additionally, promotional strategies can be introduced during low-performing months like September to boost demand.

Recommendations based on Customer dashboard:

1. Focus on High-Value Customer Segment (Age 50+)

- Introduce targeted loyalty programs to improve retention among 50+ customers.
- Offer bundled product deals to maximize revenue from this segment.
- Allocate a higher share of marketing budget toward campaigns targeting this age group.

2. Increase Purchase Frequency for Age Group 30–40

- Implement subscription models or repeat purchase incentives.
- Provide personalized recommendations based on past purchase behavior.
- Offer limited-time discounts to encourage more frequent transactions.

3. Optimize Category-Based Marketing Strategy

- Promote electronics more aggressively to customers aged 50+.
- Focus on beauty and clothing categories for younger segments (18–40).
- Design targeted campaigns aligning product categories with age-based preferences.

4. Improve Performance of Mid-Level Segment (Age 40–50)

- Launch re-engagement campaigns to increase activity in this segment.
- Provide targeted discounts and combo offers to boost both revenue and volume.
- Analyze barriers to higher spending and address them through tailored offers.

5. Implement Cross-Selling and Upselling Strategies

- Recommend complementary products across categories during purchase.
- Bundle high-performing categories to increase average order value.
- Use data-driven product suggestions to enhance customer basket size.

6. Enhance Overall Customer Retention Strategy

- Introduce tiered rewards programs based on purchase behavior.
- Use customer segmentation insights to deliver personalized marketing.
- Focus on long-term engagement rather than one-time purchases.

Recommendations based on product category analysis:

1. Prioritize High-Revenue Category (Electronics)

- Increase marketing spend on electronics to maximize revenue growth.
- Highlight premium products through targeted campaigns and promotions.
- Ensure consistent stock availability to avoid revenue loss from stockouts.

2. Leverage High-Volume Category (Clothing)

- Introduce bundle offers and discounts to increase average order value.
- Optimize pricing strategy to improve revenue without affecting demand.
- Use clothing as an entry category to attract new customers.

3. Strengthen Mid-Performing Category (Beauty)

- Improve visibility through targeted promotions and product recommendations.
- Experiment with pricing and offers to increase both volume and revenue.
- Bundle beauty products with clothing or electronics to boost sales.

4. Implement Category-Specific Marketing Strategies

- Promote electronics to older customers (50+) who show higher interest.

- Target younger segments (18–40) with clothing and beauty products.
- Personalize campaigns based on category preferences and purchase behavior.

5. Increase Average Order Value Across Categories

- Introduce cross-selling strategies (e.g., accessories with electronics).
- Offer combo deals across categories to encourage larger purchases.
- Use data-driven recommendations to suggest complementary products.

6. Optimize Product Mix and Inventory Planning

- Maintain higher inventory levels for high-demand categories like clothing.
- Focus on high-margin products within electronics to maximize profitability.
- Reduce slow-moving inventory by offering targeted discounts.

7. Improve Category Contribution Balance

- Reduce over-dependence on a single category by strengthening others.
- Invest in underperforming areas to create a more balanced revenue stream.
- Continuously monitor category performance and adjust strategy accordingly.

BUSINESS SUMMARY

Business Snapshot

- Total revenue stands at 456,000 with an average order value of 456.
- A total of 2,514 units were sold across all product categories.

Key Insights

- Electronics is the highest revenue-generating category due to higher pricing.
- Clothing drives the highest sales volume but contributes less revenue per unit.
- Customers aged 50+ contribute the most to overall revenue.

- Average order value is similar across categories, indicating consistent pricing.

What the Business Should Do

- Focus on electronics to maximize revenue growth.
- Improve revenue from clothing through pricing and bundling strategies.
- Target high-value customers (50+) with personalized marketing.

THE END